

$$\text{Output Impedance, } Z_o = \frac{R_s + h_i}{h_o R_s + h_o h_i - h_f h_r}$$

(5) Power Gain, A_p

As long as the load resistance $Z_L = R_L$ (a resistor), then:

Output power, $P_{out} = V_{out} I_{out}$ (Watts)

AC Input power, $P_{in} = V_{in} I_{in}$ (Watts)

$$\text{Power Gain, } A_p = \frac{P_{out}}{P_{in}} = \frac{V_{out} \cdot I_{out}}{V_{in} \cdot I_{in}} = A_V A_I$$

$$\text{Power Gain, } A_p = A_V A_I$$

$$\text{since } A_V = A_I \cdot \frac{Z_L}{Z_{in}}, \Rightarrow A_p = A_I \cdot \frac{Z_L}{Z_{in}} \cdot A_I$$

$$\therefore A_p = [A_I]^2 \cdot \left[\frac{Z_L}{Z_{in}} \right]$$

(6) Current gain with finite source resistance R_s , (A_{is})

When the input to the amplifier is a current source with finite resistance, it is taken into account in finding the overall current gain A_{is} . In the equivalent circuit shown in Fig. 9, the current source I_s has finite source resistance, R_s . The source current is not the transistor input current because part of it flows into R_s and part along Z_{in} .

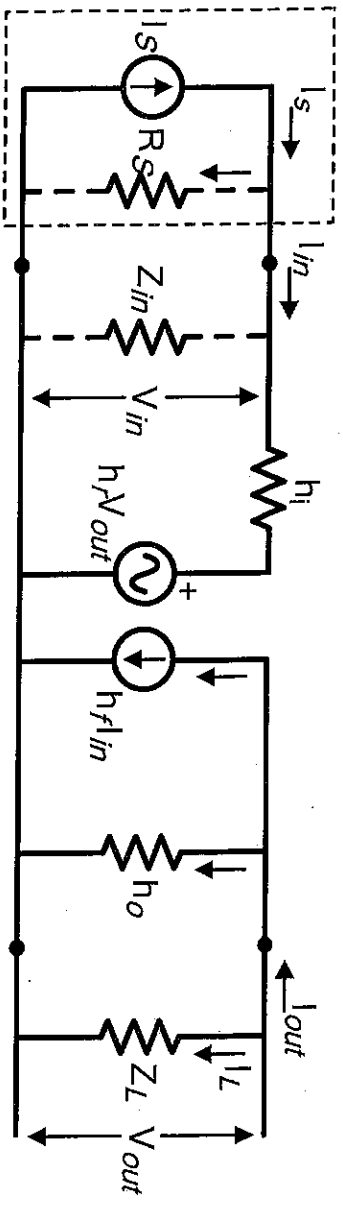


Fig.9. Equivalent circuit of transistor amplifier using h-parameters with current source having finite source resistance R_s

Overall current gain A_{is} is calculated as follows:

$$A_I = \frac{I_{out}}{I_s} = \frac{I_{out}}{I_{in}} \cdot \frac{I_{in}}{I_s} = A_I \cdot \frac{I_{in}}{I_s} \dots\dots\dots(14)$$

Applying current divider rule to the input in Fig. 9,

$$I_{in} = \frac{I_s \cdot R_s}{(R_s + Z_{in})}$$

$$\frac{I_{in}}{I_s} = \frac{R_s}{(R_s + Z_{in})}$$

Substituting it into Eqn. (14) above gives;

$$A_{IS} = A_I \cdot \frac{R_s}{(R_s + Z_{in})}$$

(7) Voltage gain with finite source resistance R_s , (A_{VS})

When the input to the amplifier is a voltage source with finite resistance, it is taken into account in finding the overall voltage gain A_{VS} . In the equivalent circuit shown in Fig. 10, the source voltage V_s has finite source resistance, R_s . Now, V_s drops over series combination of R_s and Z_{in} .

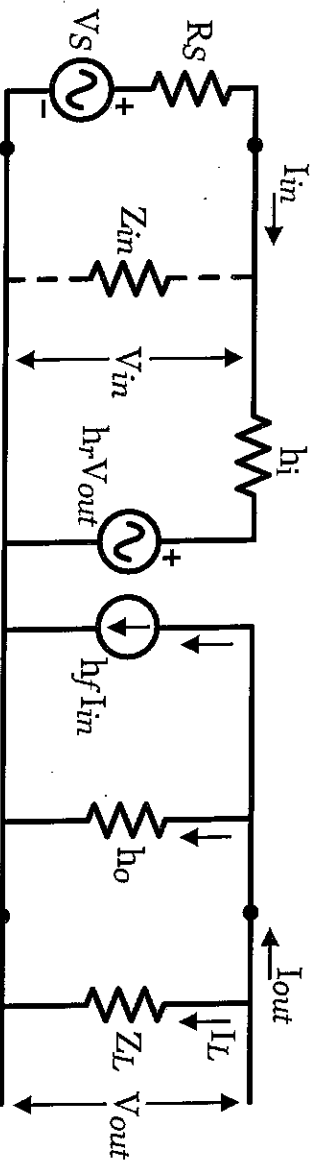


Fig.10. Equivalent circuit of transistor amplifier using h-parameters with voltage source having finite source resistance R_s

Overall voltage gain A_{VS} is calculated as follows:

$$A_{VS} = \frac{V_{out}}{V_s} = \frac{V_{out}}{V_{in}} \cdot \frac{V_{in}}{V_s} = A_V \cdot \frac{V_{in}}{V_s} \dots\dots\dots(15)$$

Applying voltage divider rule to the input in Fig. 10,

$$V_{in} = \frac{V_s \cdot Z_{in}}{(R_s + Z_{in})}$$

$$\frac{V_{in}}{V_s} = \frac{Z_{in}}{(R_s + Z_{in})}$$

Substituting it into Eqn. (15) above gives;

$$A_{VS} = A_V \cdot \frac{Z_{in}}{(R_s + Z_{in})}$$

Transistor amplifier performance can be completely assessed by the four basic relations for A_I , A_V , Z_{in} and Z_{out} as shown in Table 2 for all the transistor configurations.

Table 2. Performance of CE, CB, and CC transistor amplifiers

Parameter	CE	CB	CC
A_i	$\frac{-h_{fe}}{(1 + h_{oe}Z_L)}$	$\frac{-h_{fb}}{(1 + h_{ob}Z_L)}$	$\frac{-h_{fc}}{(1 + h_{oc}Z_L)}$
Z_i	$h_{ie} - A_{ice} \cdot h_{re} \cdot Z_L$	$h_{ib} - A_{icb} \cdot h_{rb} \cdot Z_L$	$h_{ic} - A_{icc} \cdot h_{rc} \cdot Z_L$
A_v	$\frac{Z_L}{A_{ice} \cdot Z_{ie}}$	$\frac{Z_L}{A_{icb} \cdot Z_{ib}}$	$\frac{Z_L}{A_{icc} \cdot Z_{ic}}$
Y_o	$h_{oe} - \left(\frac{h_{fe} \cdot h_{re}}{R_s + h_{ie}} \right)$	$h_{ob} - \left(\frac{h_{fb} \cdot h_{rb}}{R_s + h_{ib}} \right)$	$h_{oc} - \left(\frac{h_{fc} \cdot h_{rc}}{R_s + h_{ic}} \right)$
Merits	Inverting amplifier with reasonable voltage and current gain	Non-inverting amplifier with high Z_o	Non-inverting amplifier with large A_i . Excellent as impedance transformer

Example 1. Find the values of A_i , A_v , A_{vs} , A_{is} , Z_i , Z_o for the transistor amplifier circuit in Fig. 9. Typical values of transistor h-parameters are as follows: $h_{ie} = 1 \text{ k}\Omega$, $h_{fe} = +50$, $h_{oe} = 25 \times 10^{-6} \text{ mhos}$, $h_{re} = 4 \times 10^{-4}$

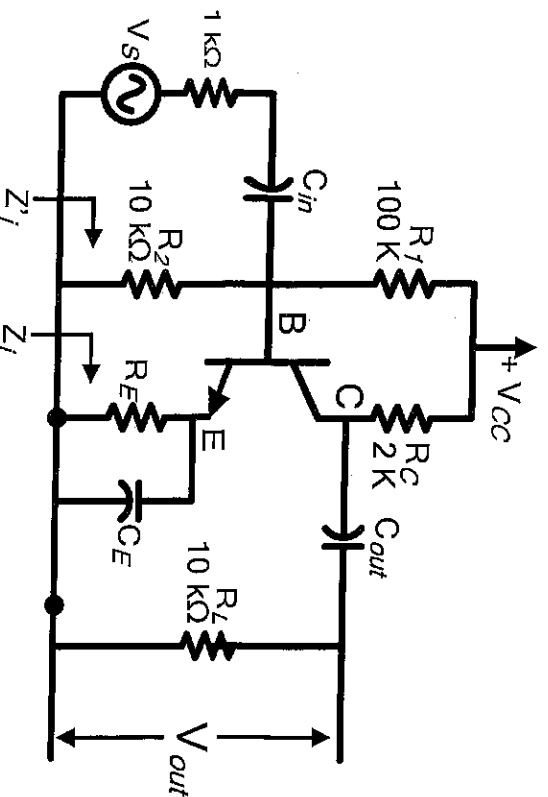


Fig. 11. Common emitter transistor amplifier circuit

Solution

First, a complete common-emitter hybrid equivalent circuit is substituted and the network redrawn as shown in Fig. 12.

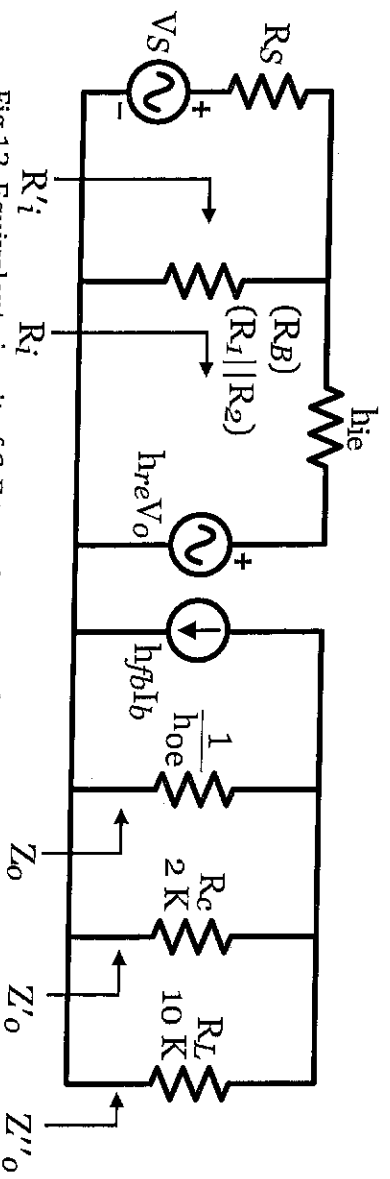


Fig. 12. Equivalent circuit of C-E transistor amplifier circuit using h-parameters